

Room 11 (P1 N=6 contour construction, P2 odd-a):
moderator summary

2026-09-26

Erdős: the direct N=4-template contour construction for N=6 fails, for a precise and now fully verified reason (Reviewer AV). Computing the exact N=6 saddle data ($x_{-3} = 0.349119 - i\pi/6$, $x_{-1} = 0.349119 + i\pi/6$, confirmed algebraically exact – $c^6 = 64$ real forces the fixed-point root w real, giving $\text{Im}(x_m) = \arg(c)/2 + \pi m/6$ as an identity, not a numerical coincidence) shows the two saddle-*pinned* constant-height contour legs coincide *exactly*, along their entire length, with two of Room 8’s five N=6 resonance loci ($\pi/6$, $5\pi/6 \equiv -\pi/6$). A calibration check (Reviewer AV, independently confirmed) shows N=4’s own saddle heights ($3\pi/8$, $-\pi/8$) miss its own three resonance loci by a clear nonzero margin – so this is a genuine N=6-specific obstruction, not an assumed pattern. The contour height is forced by the saddle equation (not a free design choice), and rerouting away from it would make the contour miss the actual stationary point, defeating the Laplace-method step that produces the leading asymptotic term (prop:qi Step 3) – a real mechanism, not assertion. **Correctly scoped conclusion (per Reviewer AV, softening the room’s own slightly stronger framing):** this rules out the *direct N=4-template transplant* specifically, for a precise, saddle-equation-forced reason; it does not prove no technique whatsoever can work at $N = 6$. Two unclosed escape routes remain explicitly open in the underlying note, one described as “not obviously possible,” the other as needing a bound not written anywhere in the repository.

Tao: pushing P2’s odd-a branch found nothing rigorous, plus an important precision caution (URGENTLY audited, resolved). No exact recursion or rigorous convergence proof was found for the odd-a branch. In the course of this, Tao’s seat found that the `ratio_amp` tool suffers catastrophic cancellation (values changing by ~ 100 orders of magnitude under $200 \rightarrow 500$ -bit precision doubling) for a *different* construction ($N = aq+2$,

which forces $N \equiv 2 \pmod{q}$, manufacturing an artificial resonance). This raised an urgent question of whether Room 7's committed findings were themselves precision-converged. **Resolved: Reviewer AU audited 10 of Room 7's actually-cited points at 200/400/800/1600-bit precision; all are bit-identical to 40 digits. Room 7's construction ($a = \text{round}(N/q)$) has no such forced resonance, so the cancellation issue does not transfer. No retraction needed; Room 7/9 stand as reported.**

Room verdict: ADVANCED. N=6's obstruction is now located with algebraic exactness (not just numerically observed), a genuine advance in precision over Room 1/8's findings, with one specific technique now precisely ruled out rather than merely stalled-on. The precision scare around Room 7 was investigated urgently and resolved negatively (no problem found) – itself valuable due diligence, not wasted effort. Neither result closes its target problem; both narrow it further with verified, not merely asserted, mechanisms.